Cosmas Engagement Engine — Findings & Decision Brief

Date: 2026-06-01 · **Prepared by:** Oscar · **Audience:** Leadership (decision) **Status:** Technical evaluation complete. Awaiting GO / NO-GO + scope.

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1. What it is

A prototype that reads a patient's **de-identified** intake/visit text, estimates the **Big Five (OCEAN)** personality traits, and maps them to one of **5 "engagement styles"** plus a **confidence level**. It is a **communication aid** — meant to help a clinician tailor *how* they talk to a patient. It is explicitly **not** a diagnosis, a psychological assessment, or a clinical instrument.

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2. What we evaluated

- **Code & fit:** reviewed the engine's code and how it would plug into our stack and DB.
- **Model swap:** the prototype ships against Anthropic Claude; we re-ran everything on **our own Azure OpenAI (gpt-4.1)** — the model we already pay for.
- **Real-data validation:** a **read-only** run over **real de-identified production transcripts** (`ai_visit_transcript_non_phi`, the dataset built for exactly this kind of analysis), plus an A/B test of one proposed fix.

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3. What we found

✅ The code is solid and the model swap works

Clean, typed, tested, honest about its limits. On the public **labeled** benchmark, gpt-4.1 reproduces literature-level accuracy (**~0.56 macro**, range 0.55–0.65 is normal for text-based Big Five). **We do not need to add Anthropic** — Azure OpenAI is enough.

Trait	Accuracy (gpt-4.1)
Extraversion	0.66
Conscientiousness	0.59

Trait	Accuracy (gpt-4.1)
Openness	0.55
Agreeableness	0.54
Neuroticism	0.48
Macro avg	**0.56**

⚠️ On our real visit transcripts, it is not usable as-is

Over a random sample of real production visits (2,000–9,500 words each):

- **Confidence was "Low" for 100% of patients** — and **not** because text was too short (evidence was rated "High" for all). The model rates almost everyone as similarly *agreeable* / *conscientious*, so it cannot cleanly separate the five styles.
- Predictions **skew heavily** toward 2 of the 5 styles (the high-agreeableness ones). A clinician would see "Low confidence" + the same couple of labels almost every time.

❌ The obvious cheap fix does not work

We hypothesized that feeding **only the patient's words** (dropping the doctor's turns) would sharpen the signal. We A/B-tested it on the same transcripts:

Style	All speakers	Patient-only
Caretakers	43%	**62%**
Guided Trusters	30%	24%
Strivers	10%	3%
Weighers	10%	7%
Free Spirits	7%	3%
Confidence	**100% Low**	**100% Low**

Patient-only made it **slightly worse** (more concentrated, still 100% Low). **Ruled out by evidence.**

Bottom line

The **integration is the easy part** (~1.5–2.5 weeks of engineering). The **value** depends on real **re-calibration for clinical text**, which is harder and uncertain. One lever remains untested: using the

intake-chatbot text (the patient's own words, in a freer register) instead of the visit transcript — we can check that quickly if desired.

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4. What needs to be decided

#	Decision	Recommendation
1	**Do we want to infer & show patient personality at all?** (Patient trust / ethics / regulatory exposure.)	Only as an **internal, opt-in pilot** — never patient-facing, never driving clinical decisions.
2	**Greenlight re-calibration, or stop here?** Wiring it in is cheap; making it *useful* is real R&D with uncertain payoff.	**Don't integrate yet.** Do a short spike: (a) test the intake-text path, (b) re-calibrate for clinical text, then re-validate and decide.
3	**LLM provider.**	Use **Azure OpenAI** (validated). No Anthropic billing.
4	**PHI handling.** We'd score the PHI transcript inside our existing pipeline — same exposure profile as today's note generation.	Acceptable, but get a quick **compliance sign-off**.
5	**Per-provider vs global.** The engine's rubric is global; our standing rule is per-provider customization with no cross-provider migration.	**Global for v1**, revisit later.

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5. One-liner for the discussion

> *"*The model we already pay for can do this, and wiring it in is cheap — but on real visits it currently shows 'Low confidence' for every patient, and the cheap fix didn't help. Recommendation: a short re-calibration spike before we commit to a full build, not a green light to integrate."*

Methodology note: all numbers above come from runs against our Azure OpenAI gpt-4.1 deployment. The accuracy figures use the public Pennebaker & King essays corpus (human-labeled). The production findings use a read-only sample of de-identified transcripts. Full data tables and reproduction scripts are documented in the engineering notes (`2026-06-01-engagement-engine-decision.md` *).*